

A Norming-Face Obstruction for L^p State Spaces

Automatic Math Discovery Smoke Test

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Source problem

The source paper asks for an analogue, for $1 < p < \infty$, of its characterization of weakly compact state spaces in $L^1(\mu, X)$.

not a very smooth point (see, [11, Proposition 2.3]).

3.2. Open Problems: In what follows, we present some open problems.

Problem 3.17. *In Theorems 2.5 and 2.8, we studied the w^* - w and w^* - $\|\cdot\|$ points of continuity of the state spaces. However, a characterization of w^* - w points of continuity of the dual unit ball $\ell^q(X^*)_1$ is not known. In particular, even for a norm-attaining functional $(x_i^*) \in \ell^q(X^*)$, the conclusion remains open.*

Problem 3.18. *In Theorem 3.6, we have presented a characterization of weakly compact state spaces in the space $L^1(\mu, X)$. A similar result is not known for the space $L^p(\mu, X)$, where $1 < p < \infty$.*

Result

This packet gives a necessary condition for weak compactness of the state space of a constant function. It supplies a simple non-reflexive obstruction which is not covered by the reflexive-domain partial answer.

Throughout this note the scalar field is real.

Theorem 1. *Let (Ω, Σ, μ) be a probability space, let $1 < p < \infty$, and let q be conjugate to p . Let X be a Banach space, let $x \in S_X$, and put $f(\omega) = x$ for all $\omega \in \Omega$. Define*

$$S_f = \left\{ g \in B_{L^q(\mu, X^*)} : \int_{\Omega} g(\omega)(x) d\mu(\omega) = 1 \right\}$$

and

$$F_x = \{x^* \in B_{X^*} : x^*(x) = 1\}.$$

If S_f is weakly compact in $L^q(\mu, X^)$, then F_x is weakly compact in X^* .*

Proof. For each $x^* \in F_x$, let Jx^* be the constant function with value x^* . Since μ is a probability measure,

$$\|Jx^*\|_{L^q(\mu, X^*)} = \|x^*\| \leq 1$$

and

$$\int_{\Omega} (Jx^*)(\omega)(x) d\mu(\omega) = x^*(x) = 1.$$

Thus $J(F_x) \subset S_f$.

Now define the bounded linear operator

$$Q : L^q(\mu, X^*) \longrightarrow X^*, \quad Q(g) = \int_{\Omega} g(\omega) d\mu(\omega),$$

where the integral is the Bochner integral. Holder's inequality gives $\|Q(g)\| \leq \|g\|_{L^q(\mu, X^*)}$, and $Q(Jx^*) = x^*$ for every $x^* \in F_x$.

If S_f is weakly compact, then $J(F_x)$ is relatively weakly compact as a subset of S_f . Since Q is weak-to-weak continuous, $Q(J(F_x)) = F_x$ is relatively weakly compact in X^* . Finally, F_x is weakly closed: it is the intersection of the weakly closed unit ball B_{X^*} with the weakly closed affine hyperplane $\{x^* : x^*(x) = 1\}$. Therefore F_x is weakly compact. $\square$

Corollary 1. *If $x \in S_X$ has a non-weakly-compact norming face F_x , then the state space of the constant function $f(\omega) = x$ is not weakly compact in $L^q(\mu, X^*)$.*

Example 1. *Take $X = \ell_1$ and $x = e_1$. Then $X^* = \ell_{\infty}$ and*

$$F_{e_1} = \{a = (a_n) \in B_{\ell_{\infty}} : a_1 = 1\}.$$

This face is not weakly compact. Indeed, translating by $(-1, 0, 0, \dots)$ identifies it with $\{0\} \times B_{\ell_{\infty}}$, which is isometric to the closed unit ball of ℓ_{∞} . If F_{e_1} were weakly compact, then $B_{\ell_{\infty}}$ would be weakly compact, forcing ℓ_{∞} to be reflexive, a contradiction.

Hence, for any probability space and every $1 < p < \infty$, the state space of the constant e_1 function in $L^p(\mu, \ell_1)$ is not weakly compact.

Verification code

The script `code/check_constant_face_obstruction.py` verifies a finite-truncation version of the constant-face inclusion: for several vectors $a \in B_{\ell_{\infty}^n}$ with first coordinate 1, the constant function has L^q norm 1 and pairing 1 with the constant e_1 function.

Scope

This is a partial answer to Problem 3.18. It does not characterize all weakly compact L^p state spaces. It says that, even for constant functions, weak compactness of the state space detects weak compactness of the corresponding norming face in X^* .